

Quick Check

Lesson 4: Quadrat Sampling: Is the Distribution of Grass and Weeds Uniform Across the School Field?

7 multiple-choice questions • choose A, B, C or D

Sub-Strand 1.2: Ecology | Lesson 4

A team is sampling the grass and weeds in a zone of the school field using a 1 m x 1 m quadrat. Why must they place the quadrat at random locations instead of where the grass looks nicest?

- A** So that every quadrat contains exactly the same number of plants
- B** So that the sample is not biased and represents the whole zone
- C** So that the counting is finished faster
- D** So that only the most common weed is counted

A team places a 1 m x 1 m quadrat at 5 random spots in a sunny zone of the school field. The numbers of weed plants counted are 12, 8, 10, 15 and 5. What is the density of weeds in plants per m²?

A 10 plants per m²

B 5 plants per m²

C 15 plants per m²

D 50 plants per m²

In a damp zone of the school field, the mean number of grass tufts is 6 per 1 m x 1 m quadrat. The zone has a total area of 200 m². What is the estimated total number of grass tufts in the zone?

- A 33 tufts
- B 1200 tufts
- C 120 tufts
- D 206 tufts

A class uses quadrats to estimate that there are 1200 weed plants in a zone. Why is this number only an estimate?

- A** Weeds cannot be counted accurately because they are too small
- B** Quadrats can only be used on grass and not on weeds
- C** The area of the quadrat was not known
- D** Only a small part of the zone was sampled and the result is assumed to represent the whole zone

Home groups compare the mean plant density of two zones on the school field: 42 plants per m^2 in a damp zone near a tap and 6 plants per m^2 in a heavily trampled zone. Which claim is best supported by this evidence?

- A** Trampling has been proved to be the only cause of the low density
- B** The distribution of plants is not uniform, because density was much higher in the damp zone than in the trampled zone
- C** The damp zone is healthier because more plants always means a healthier ecosystem
- D** The distribution of plants is uniform, because plants grow everywhere on the field

Along a Kenyan river, aquatic weeds and algae grow thickly in one stretch where fertiliser and sewage from nearby farms enter the water. Fish are fewer there. Which explanation fits best?

- A** Fish are fewer only because they are old and have no food
- B** Extra nutrients let algae and weeds boom, which can reduce oxygen in the water and harm the fish
- C** More weeds means more food and oxygen, so the fish there are healthier
- D** The weeds grow thickly only because the water is the same everywhere along the river

On the school field, plant density is much lower on a heavily used footpath than beside a tap. Which is the best explanation?

- A** Trampling and compacted soil damage plants and limit root growth, while the soil beside the tap has more water
- B** Plants need no water, so the tap makes no difference to density
- C** Seeds cannot grow anywhere that the soil is wet
- D** Plants are spread evenly, so the difference must be due only to counting mistakes

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).